Banega — Functional Requirements Document

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1. Overview

Banega answers "aaj khaane mein kya hai?" for a whole household. It stores every family member's food rules once, then suggests recipes for each meal that the most people can eat. Suggestions come from an AI assistant called Maharaj.

Problem. Indian homes decide three meals a day across mixed diets (Jain, veg, egg, non-veg), spice levels and health needs. The decision usually happens between the person cooking (family, cook or house help) and whoever is home, from memory.

Goals for v1

- Onboard a household and every member's preferences in under 3 minutes.
- Return 3–6 meal options per query, each labelled with who can eat it and what to tweak.
- Turn chosen dishes into a daily cooking ticket for the cook, with required items checked against kitchen stock.

Reference prototype. The clickable design on the Banega canvas (artboard "App prototype") is the source of truth for layout, copy and motion. Its recipe list and AI replies are hard-coded; this document defines the real behaviour behind them.

2. Users and roles

One household account holds one admin, many member profiles and optional cook access.

Role	Logs in?	Can do
Household admin	Yes (phone OTP)	Create household, add/edit/remove members, ask Maharaj, Order staples, recieves messages on whatsapp from the app / maharaj on low inventory etc, build menu, send to cook, invite others

Role	Logs in?	Can do
Family member (adult)	Optional, via invite	Edit own preferences, ask Maharaj, add dishes to the day menu
Profile-only member (kids, elders)	No	Exists as a preference profile managed by the admin
Cook (hired, or a family member such as mom)	Yes. Hired cook: own OTP login, cook mode only. Family cook: same login, switches between family and cook mode	Gets a daily cooking ticket, opens recipes, marks items out of stock, requests ration, closes tickets, sees ratings

Cooking is a role, not a person. In most homes the person cooking is mom or another family member; some homes have a hired cook, and many mix both. Onboarding asks "Ghar mein khaana kaun banata hai?" and sets one of these setups:

Setup	Who holds the cook role	How the app behaves
Family cooks	One family member (e.g. Mummy)	No separate cook login; tickets appear as "Aaj ka plan" in her own app; no accept step; she approves her own substitutes and ration list
Hired cook	Maharaj / didi with their own phone	Separate Hindi-first cook mode; tickets need accepting; dish changes need admin approval
Both	Hired cook for some meals, family for others	Each ticket is assigned per meal (e.g. cook does lunch, Mummy does dinner)
Rotates	Several family members take turns	Each ticket has an assignee chosen when the menu is confirmed

The setup can be changed any time in settings.

3. Scope

v1 is a mobile app (iOS + Android) covering onboarding, recommendations, chat, recipe detail and the day menu. Everything else waits.

Area	v1 (MVP)	Later
Onboarding	Household + member profiles, diet, spice, needs, cuisines	Import from contacts, member photos
Recommendations	Per-meal ranked list with family-fit labels	Weekly planner, leftovers, seasonal and festival menus
Maharaj chat	Text queries in English, Hindi and Hinglish	Voice input, image of fridge/pantry
Recipe source	1,000+ curated home recipes covering all Indian regional cuisines	User-submitted family recipes
Menu	Day menu becomes a cooking ticket for the cook; kitchen stock, ration requests, AI substitutes, ratings	Barcode/bill scan for stock, cook attendance and salary, multiple cooks per household
Commerce	"Order missing items" as a bill-style list; admin orders manually	Grocery partner integration and in-app checkout

Pricing. 7-day free trial, then ₹50 per week for all v1 features, billed as an auto-renewing weekly subscription through Google Play and the App Store. Plans and prices will change as features are added.

4. Screens and flow

First launch runs Welcome → Family → Preferences → Loading → Home. Every later launch opens on Home.

#	Screen	Purpose	Main actions	Goes to
S1	Welcome	Brand intro with the 3D thali	Set up household; view demo family	S2, S4
S2	Family ("Ghar mein kaun kaun hai?")	List of member cards with a one-line summary	Tap member; add member; continue	S3, S5
S3	Preferences	Edit one member: diet, spice, needs, cuisines	Switch member tab; save	S2
S5	Loading	Shown while the first recommendations are computed (max 3 s)	None	S4

#	Screen	Purpose	Main actions	Goes to
S4	Home	Meal-slot tabs, Maharaj's pick, quick asks, recipe grid, menu bar, bottom nav	Search/ask; switch slot; ADD; open recipe	S6, S7, S8, S2
S6	Maharaj chat	Conversation with recipe cards and follow-up chips	Send text; tap chip; ADD; open recipe	S7, S8
S7	Recipe detail	Photo, who can eat it, ingredients, steps	Add to menu; open video recipe	Back to origin
S8	Day menu	Chosen dishes, merged grocery list	Remove dish; send to cook; order missing items	S4

Back navigation from S7 returns to whichever screen opened it (Home, Chat or Menu).

Cook app screens. A hired cook lands straight in cook mode with Hindi-first copy, large type and icons. A family cook sees the same screens as an "Aaj ka plan" tab in her normal app, in her chosen language, without the accept and approval steps.

#	Screen	Purpose	Main actions	Goes to
K1	Aaj ke tickets	Today's cooking tickets by meal, with status and due time	Open ticket	K2
K2	Ticket detail	Dishes, servings, who is eating, per-member tweaks, required items with stock status	Accept; start cooking; mark item out of stock; open recipe; complete	K3, K4, K5
K3	Recipe (cook view)	Steps and ingredients scaled to today's servings, with an audio read-out	Back	K2
K4	Stock issue	AI's alternate recipe or replacement dish for a missing item	Accept suggestion; ask family	K2
K5	Rasoi stock	Household staples by category with in-stock / low / out	Update levels; request ration	K6

#	Screen	Purpose	Main actions	Goes to
K6	Ration request	Items to buy with quantities, pre-filled from low stock and upcoming tickets	Send to admin	K5
K7	Meri ratings	Ratings and comments per dish, weekly average	View details	—

Family-side additions: a ticket status card on Home, a ration approval screen, and a rate-the-meal sheet shown after a ticket closes.

5. Functional requirements

Each requirement has an ID for tickets and QA. "Must" items block v1 release; "Should" items can slip one sprint.

5.1 Account and household

ID	Priority	Requirement	Acceptance criteria
FR-A1	Must	Sign up and log in with Indian mobile number + OTP	OTP arrives in under 30 s; 3 wrong attempts lock for 10 min
FR-A2	Must	Create one household with a name ("Purohit Parivaar")	Name shows in the Home header
FR-A3	Should	Invite an adult member by WhatsApp link	Invitee lands logged in to the same household
FR-A4	Must	"View demo family" skips setup with 5 sample members	Demo data is clearly labelled and can be cleared
FR-A5	Must	Onboarding asks who cooks: a family member, a hired cook, both, or several people in turns	Answer sets the cook setup (section 2); a family member can be picked as cook without a second phone
FR-A6	Must	7-day free trial from household creation, then a ₹50/week subscription paid by the admin; one subscription covers the whole household including the cook	Reminder on day 5 and day 7; after expiry the app is read-only until the admin subscribes

5.2 Member profiles

ID	Priority	Requirement	Acceptance criteria
FR-M1	Must	Add, rename and remove members (max 12)	New member defaults to Veg, Medium spice, no needs
FR-M2	Must	Diet: exactly one of Veg, Jain, Egg, Non-veg	Single-select; change saves instantly
FR-M3	Must	Spice: exactly one of Kam, Medium, Teekha	Segmented control
FR-M4	Must	Needs (multi-select): Low oil, Low sugar, High protein, No maida, Kid-friendly	Any combination, including none
FR-M5	Must	Favourite cuisines (multi-select), grouped by region as in section 7.4, plus Indo-Chinese and Continental	Any combination
FR-M6	Should	Free-text dislikes and allergies (e.g. "no baingan", "peanut allergy")	Allergy items become hard exclusions in the matching engine
FR-M7	Must	Member card shows diet badge + one-line summary	Summary never wraps; truncates with ellipsis

5.3 Home

ID	Priority	Requirement	Acceptance criteria
FR-H1	Must	Default meal slot by device time: Nashta before 11:00, Lunch 11:00–16:00, Dinner after 16:00	User can switch slot; choice persists for the session
FR-H2	Must	"Maharaj ka pick" banner shows the top-ranked dish for the slot	Tapping opens Recipe detail
FR-H3	Must	Recipe grid shows 8 top-ranked dishes for the slot	Each tile: photo, veg/egg/non-veg mark, time, name, cuisine, fit badge, ADD
FR-H4	Must	Quick-ask chips send a preset query to Maharaj	Opens Chat with the query already sent

ID	Priority	Requirement	Acceptance criteria
FR-H5	Must	Search bar opens Chat; placeholder rotates example queries	Rotation every 2.5 s
FR-H6	Must	Sticky menu bar appears once one dish is added	Shows count and names; tap opens Day menu
FR-H7	Must	Header shows household name and key constraints (e.g. "Dadi (Jain) · Papa (low sugar)")	Max 3 constraints shown

5.4 Maharaj chat

ID	Priority	Requirement	Acceptance criteria
FR-C1	Must	Accept free text in English, Hindi (Devanagari) and Hinglish	Same query in all three returns the same intent
FR-C2	Must	Show a typing state while the answer is prepared	Appears within 200 ms of send
FR-C3	Must	Reply = one short message + 3–4 recipe cards + 2–4 follow-up chips	First token in under 2 s, full reply in under 6 s (p90)
FR-C4	Must	Each card: photo, mark, time, name, fit badge, one-line note, ADD	Note names the main tweak or who can't eat it
FR-C5	Must	Keep conversation context within a session ("aur halka?" refines the last answer)	Follow-up narrows the previous result set
FR-C6	Must	When nothing matches, say so and show the closest options	Never an empty reply
FR-C7	Should	Chat history saved for 30 days	Visible when Chat reopens

5.5 Recipe detail

ID	Priority	Requirement	Acceptance criteria
FR-R1	Must	Hero photo, diet label, cuisine, total time, name, one-line description	Photo loads under 1 s on 4G

ID	Priority	Requirement	Acceptance criteria
FR-R2	Must	"Kaun khayega": every member with Perfect / Tweak / Nahi and a reason	Uses the matching engine (section 6)
FR-R3	Must	Ingredients list with quantities, scaled to household size	Scaling rounds to kitchen units (cup, tsp, pieces)
FR-R4	Must	Numbered steps	3–10 steps
FR-R5	Must	"Video recipe" opens the source video or page	Opens in-app browser
FR-R6	Must	Add to / remove from day menu	Button state mirrors the menu

5.6 Day menu and sharing

ID	Priority	Requirement	Acceptance criteria
FR-P1	Must	One menu per household per meal slot per date	Adding the same dish twice is ignored
FR-P2	Must	Coverage line: whether every member has at least one dish they can eat	Names the uncovered members when false
FR-P3	Must	Merged grocery list with duplicates combined	Checkboxes persist per menu
FR-P4	Must	"Send to cook" turns the menu into a cooking ticket in the cook app (WhatsApp message only if the cook has no app)	Message readable without the app; Hindi option
FR-P5	Should	"Order missing items" generates a bill-style list of unchecked items (item, quantity, unit); the admin orders manually from any shop or app	List can be shared on WhatsApp, saved as image or copied

5.7 Cook app and daily tickets

Every confirmed menu becomes one ticket per meal. The cook works only from tickets.

ID	Priority	Requirement	Acceptance criteria
FR-K1	Must	Admin gives the cook role to a family member or to a hired cook (name + phone); a hired cook logs in with OTP and sees only cook mode, a family cook switches modes	Cook cannot edit member profiles or other households' data
FR-K2	Must	A ticket is created when the family confirms a meal's menu	Holds date, meal slot, dishes, servings, eaters, due time, notes
FR-K3	Must	Cook gets a push notification for each new or changed ticket	Arrives within 10 s; ticket shows "Updated" tag with what changed
FR-K4	Must	Ticket lists per-member tweaks (e.g. "Dadi: bina pyaaz-lahsun katori alag")	Pulled from the matching engine
FR-K5	Must	Cook opens any dish's recipe, scaled to the ticket's servings	Includes audio read-out in Hindi
FR-K6	Must	Cook can message the family on the ticket ("dahi khatam hai, kya karoon?")	Family sees it on the Home ticket card
FR-K7	Must	Admin can edit or cancel a ticket until status is Cooking	Later edits need cook acceptance

Ticket states

State	Set by	Next states
New	System when menu is confirmed	Accepted, Cancelled
Accepted	Cook	Stock issue, Cooking, Cancelled
Stock issue	Cook marks a required item out of stock	Accepted (after a substitute is approved)
Cooking	Cook	Ready
Ready	Cook	Completed
Completed	Cook or admin after serving	Rated (after ratings come in)
Cancelled	Admin	—

Cook availability (On duty / Off duty toggle)

Every person with the cook role, hired or family, has an On duty / Off duty switch at the top of cook mode. Turning it off marks them logged out of the kitchen for that period.

ID	Priority	Requirement	Acceptance criteria
FR-K8	Must	Cook toggles On duty / Off duty, with an optional reason (chhutti, beemar, bahar gaye) and return date/time	Status shows on the family Home screen within 10 s
FR-K9	Must	Off duty can be set now or scheduled ahead (e.g. Sunday off every week, leave from 12–15 Oct)	Recurring and one-off schedules both supported
FR-K10	Must	While Off duty, the cook gets no new tickets or push notifications	Tickets already accepted stay with them unless reassigned
FR-K11	Must	Open and upcoming tickets for the off-duty period move to "Unassigned" and the admin is alerted	Admin reassigns to another cook-role member in one tap
FR-K12	Must	If nobody else can cook, Maharaj suggests easy options: 15-minute dishes, ready-to-cook items in stock, or ordering in	Suggestions respect everyone's diet rules
FR-K13	Should	Off-duty history is visible to the admin as a simple monthly calendar	For hired cooks, supports attendance and salary later

Turning back On duty restores notifications and shows tickets assigned for the rest of the day.

Primary cook, backup cook and "Chef for the day"

The household names one primary cook and optionally one backup. Anyone else in the family (mom, father, a teenager) can also step in for a day.

ID	Priority	Requirement	Acceptance criteria
FR-K14	Must	Admin sets a primary cook and an optional backup cook from the household's members or hired cook	When the primary goes Off duty, tickets go to the backup first, then to Unassigned

ID	Priority	Requirement	Acceptance criteria
FR-K15	Must	Any member with the app can switch on "Aaj main chef hoon" (Chef for the day) for today, or for chosen meals	They get cook mode for those tickets; the regular cook is told they are free for those meals
FR-K16	Must	Only one person holds each ticket; if two people claim the same meal, the first claim wins and the second sees who has it	Ticket shows the current chef's name and avatar
FR-K17	Must	Chef for the day ends automatically at midnight or when they switch it off	Unfinished tickets return to the primary or backup cook
FR-K18	Should	Ratings for that meal go to whoever cooked it	Shown in their own "Meri ratings", even if they are not the usual cook

5.8 Kitchen stock and ration

The app keeps a live list of the household's staples (the "Rasoi stock"). Every ticket checks it and builds a list of required items. The full starter catalogue of about 110 items is in the **Kitchen staples** tab.

Full catalogue: Kitchen staples

ID	Priority	Requirement	Acceptance criteria
FR-S1	Must	New households start with the staples catalogue; admin or cook removes what they never use	Catalogue is grouped by category, Hindi + English names
FR-S2	Must	Each item has a stock level: In stock / Low / Out, with an optional quantity and unit	Cook changes level in one tap
FR-S3	Must	Each ticket shows "Required items": every ingredient of every dish, merged, with quantity for the servings	Each line shows stock status; Out items are highlighted
FR-S4	Must	Completing a ticket lowers stock estimates for used items	Items crossing a threshold turn Low automatically

ID	Priority	Requirement	Acceptance criteria
FR-S5	Must	Cook raises a ration request: pre-filled with Low and Out items plus items needed for tickets in the next 48 h	Cook can edit items and quantities before sending
FR-S6	Must	Admin approves, edits or rejects a ration request	Approved request becomes a bill-style list; admin orders manually, then marks it "Bought"
FR-S7	Must	Marking a request Bought sets its items back to In stock	Cook is notified
FR-S8	Should	Weekly ration suggestion based on the week's planned tickets	Shown every Sunday morning

5.9 Out-of-stock handling

When a required item is Out, Maharaj tries to save the same dish first, and only then suggests a different one. The family always approves a dish change.

1. **Trigger.** The cook marks an item Out on a ticket, or the stock check at ticket creation finds one.
2. **Same dish, different recipe.** Maharaj searches for an alternate recipe of the same dish that avoids the missing item, using only In-stock items (e.g. no dahi for kadhi → kadhi with buttermilk from milk + lemon). The result must still pass the matching engine for everyone who could eat the original.
3. **Substitute ingredient.** If no alternate recipe exists, suggest a safe swap (e.g. tomato puree for fresh tomatoes) with the change in quantity.
4. **Different dish.** If neither works, suggest up to 3 replacement dishes for the same slot that are fully in stock and rank highest for the family.
5. **Approval.** Steps 2–3 can be accepted by the cook alone if the admin allowed it in settings; step 4 always goes to the admin, who picks a dish or lets the cook choose.
6. **Timeout.** If nobody answers within 30 min of the due time, the cook may use the top suggestion; the ticket records who approved what.

ID	Priority	Requirement	Acceptance criteria
FR-O1	Must	Suggestions appear within 5 s of an item being marked Out	Shows which step (2, 3 or 4) the suggestion came from

ID	Priority	Requirement	Acceptance criteria
FR-O2	Must	No suggestion may break a member's No rules (Jain, veg, allergy)	Enforced by the matching engine, not the LLM
FR-O3	Must	Accepted changes update the ticket, required items and the family's view	Change log visible on the ticket
FR-O4	Should	The missing item is added to the next ration request automatically	Cook can remove it

5.10 Ratings and learning

After a ticket is Completed, everyone who ate rates the food; the cook sees every rating, and Maharaj learns from it. Rewards for the cook are out of v1.

ID	Priority	Requirement	Acceptance criteria
FR-F1	Must	30 min after Completed, each eater with the app gets a "Khaana kaisa laga?" prompt	One prompt per meal, not per dish
FR-F2	Must	Rating = 1–5 per dish, optional quick tags (Perfect namak, Zyada teekha, Kam pakka, Bahut tasty) and an optional note	Admin can rate on behalf of profile-only members (kids, elders)
FR-F3	Must	Cook sees each rating with the member's name, tags and note on the ticket and in Meri ratings	Visible within 1 min of submission
FR-F4	Must	Weekly average and top-rated dishes shown to cook and admin	Updated every Monday
FR-F5	Must	Maharaj learns from meals: ratings, tags and what was actually cooked (including substitutes) re-rank future suggestions per household and per member	A dish rated 4–5 by a member ranks higher for them; 1–2 lowers it for 30 days; repeated tags propose a tweak (e.g. "Papa: kam namak") that the admin confirms
FR-F6	Must	Cooked history is kept per household	Maharaj avoids repeating the same dish within 3 days unless asked

When the cook is a family member, the rating prompt leads with appreciation tags (Bahut tasty, Perfect namak) and asks "kya badlein agli baar?" instead of showing stars first. She can turn ratings off from her own settings.

6. Family-fit matching engine

For every (member, recipe) pair the engine returns Fit, Tweak or No, plus one reason. Rankings and every badge in the app come from these results. Run it server-side and deterministically; the AI never overrides a No.

6.1 Per-member rules (checked in this order; the first No wins)

Rule	Condition	Result	Reason shown
Jain	Member is Jain and recipe is not Jain-safe	No	Pyaaz-lahsun hai
Veg	Member is Veg and recipe contains egg or meat	No	Isme anda hai / Non-veg hai
Egg	Member is Egg and recipe contains meat or fish	No	Non-veg hai
No maida	Member has No maida and recipe uses maida	No	Maida hai
Allergy	Recipe contains a listed allergen	No	Contains <allergen>
Spice down	Member is Kam and recipe spice = 3	Tweak	Unki katori mein kam mirch
Spice up	Member is Teekha and recipe spice = 1	Tweak	Upar se laal mirch / achaar
Low sugar	Recipe not tagged low-sugar	Tweak	Portion chhota, salad zyada
Low oil	Recipe not tagged low-oil	Tweak	Kam tel mein banao
High protein	Recipe not tagged protein	Tweak	Saath mein dahi/paneer
Kid-friendly	Recipe not kid-tagged and spice > 1	Tweak	Bachche ke liye alag, halka
Otherwise	—	Fit	Bilkul sahi / Favourite cuisine bhi

When several Tweaks apply, show the first; store all.

6.2 Recipe score and badge

Score = 10 × members who can eat it (Fit or Tweak) + 3 × Fit members + 2 × members whose favourite cuisine matches − 15 if anyone is No. Ties break on shorter cooking time.

Badge	Condition	Colour token
Sabke liye	Everyone is Fit	fit-soft
Sab · N tweak	Everyone can eat, N need a tweak	tweak-soft
X/Y ke liye	At least one member is No	no-soft

6.3 Recipe tags the engine needs

Every recipe must carry: diet (veg / egg / nonveg), jain_safe (bool), spice (1–3), meal slots (nashta / lunch / dinner), cuisine, cook time in minutes, and tags from this set: low_oil, low_sugar, protein, kid, light, rainy, special, maida. Tagging is done by the content team with AI assistance and human review.

7. Maharaj AI assistant

Maharaj is an LLM that turns the user's words into structured filters, lets the matching engine pick recipes, then writes a short Hinglish reply. The LLM never invents recipes, ingredients or nutrition claims.

7.1 Request pipeline

1. **Parse.** Send the query, the last 6 chat turns, the household profile and the current slot to the LLM. It returns JSON only: `{slot, filters[], cuisines[], member_focus, max_minutes, keywords[]}`.
2. **Retrieve.** Query the recipe database with those filters (hard filters first, then keyword and embedding search). If fewer than 3 results, drop soft filters one at a time and note which were dropped.
3. **Rank.** Run the matching engine (section 6) on the results; take the top 4.
4. **Write.** The LLM writes one message of at most 2 sentences plus 2–4 follow-up chips, using only the ranked results and their fit data.
5. **Return** the message, recipe IDs, fit summaries and chips to the app.

7.2 Intents the parser must recognise

User says (examples)	Filter produced
nashta, breakfast, subah / lunch, dopahar / dinner, raat	slot
halka, light, bimaar, simple	tag: light
jaldi, quick, 15 min, fast	max_minutes: 20
protein, gym	tag: protein
mehmaan, guest, party, special	tag: special
baarish, rain	tag: rainy
tiffin, bachchon ke liye, beti/daughter	tag: kid
paneer, dal, any ingredient	keyword
any region or cuisine name (kerala, kashmiri, chettinad, gujarati, bengali, chinese…)	cuisine
A member's name ("Dadi ke liye")	member_focus: only recipes where that member is not No

7.3 Reply rules

- Hinglish by default; switch to the language of the query when it is clearly English or Hindi.
- The first line states the count and the filters applied (e.g. "Dinner ke liye 4 options — halka").
- The second line names how many options work for the whole family and the key constraints respected.
- No medical advice. "Low sugar" and similar are preference tags, not health claims.

7.4 Recipe sourcing

- v1 ships with at least 1,000 everyday home recipes covering every region of India, each with photo, tags, ingredients mapped to staples, and steps.
- Recipes are gathered from publicly available sources. The content team rewrites steps in Banega's own words and format, uses photos that are shot, generated or licensed, and stores and credits the source link on Recipe detail.
- Do not scrape sites whose terms forbid it. The "Video recipe" link must point to the recipe's own source.

Coverage target for the first 1,000 recipes. Weighted toward what Indian homes cook daily: dals, sabzis, rotis, rice, breakfasts. Festival and restaurant-style dishes stay under 10%.

Region	Cuisines covered	Target recipes
North	Punjabi, Delhi/UP, Awadhi, Kashmiri, Himachali, Uttarakhandi	200
West	Rajasthani, Gujarati, Maharashtrian, Goan, Sindhi, Parsi	200
South	Tamil (incl. Chettinad), Kerala, Karnataka (incl. Udupi, Mangalorean), Andhra, Telangana/Hyderabadi	220
East	Bengali, Odia, Bihari, Jharkhandi	130
North-East	Assamese, Manipuri, Naga, Meghalayan, Sikkimese	60
Central	Madhya Pradesh, Chhattisgarhi	50
Pan-Indian everyday	Khichdi, dal-chawal, parathas, chillas, tiffin items, kids' food	100
Indo-Chinese and Continental home-style	Hakka noodles, fried rice, pasta, sandwiches	40
Total		**1,000**

Every recipe must carry the section 6.3 tags, including jain_safe, so each region has Jain, veg, egg and non-veg options where the cuisine allows.

8. Data model

Fifteen core entities (no Reward entity in v1). IDs are UUIDs; all tables carry created_at and updated_at.

Entity	Key fields	Relations
User	phone, name, language (en / hi / hinglish), role (admin / member / cook)	belongs to one Household

Entity	Key fields	Relations
Household	name, city, default_servings, cook_setup (family / hired / both / rotates), cook_can_approve_swaps (bool)	has many Members, Cooks, Menus, Tickets, PantryItems
Subscription	household_id, store (play / app_store), status (trial / active / expired), trial_ends_at, renews_at, price_inr (50/week)	belongs to Household
Member	name, color, diet (veg / jain / egg / nonveg), spice (kam / medium / teekha), needs[], cuisines[], allergies[], dislikes[], is_kid, linked_user_id?	belongs to Household
Cook	member_id or user_id, is_hired, rank (primary / backup / none), duty_status (on / off), off_duty_schedule[], chef_for_day_until?, phone, languages[]	belongs to Household
Recipe	name, subtitle, photo_url, region, cuisine, diet, jain_safe, spice (1–3), meals[], minutes, tags[], ingredients[{staple_id, qty, unit}], steps[], alt_recipe_ids[], source_url, source_credit	many-to-many with Menus
Menu	household_id, date, slot (nashta / lunch / dinner), recipe_ids[], confirmed_at	belongs to Household; creates one Ticket
Ticket	menu_id, assignee_id (primary, backup or chef for the day), status, due_at, servings, eaters[], dishes[{recipe_id, tweaks[]}], required_items[{staple_id, qty, unit, stock_status}], change_log[], messages[]	belongs to Menu
Staple	name_en, name_hi, category, default_unit, perishable (bool)	global catalogue

Entity	Key fields	Relations
PantryItem	household_id, staple_id, level (in / low / out), qty?, unit?, updated_by, updated_at	belongs to Household
RationRequest	household_id, raised_by, items[{staple_id, qty, unit}], status (draft / sent / approved / rejected / bought), approved_by	belongs to Household; renders as a bill-style list
Rating	ticket_id, recipe_id, member_id, cook_id, stars (1–5), tags[], note	belongs to Ticket
CookedHistory	household_id, date, slot, recipe_id, substitutes_used[], avg_rating	feeds Maharaj's learning (FR-F5, FR-F6)
ChatSession, FitCache	chat messages; per member × recipe fit result and reason	belong to Household

9. APIs and integrations

The app talks to one REST backend; the backend talks to the LLM, recipe sources and WhatsApp.

9.1 App → backend endpoints

Method	Path	Purpose
POST	/auth/otp, /auth/verify	Phone login (admin, member or cook)
GET / POST	/subscription, /subscription/verify	Trial status; validate store purchase
GET / PUT	/household	Read or rename household, cook setup and permissions
GET / POST / PATCH / DELETE	/members, /members/{id}	Manage member profiles
GET / POST / PATCH / DELETE	/cooks, /cooks/{id}	Add cooks, set primary/backup, duty status and schedule
POST / DELETE	/cooks/chef-for-day	Claim or release today's meals as Chef for the day

Method	Path	Purpose
GET	/recommendations?slot=&date=	Ranked recipes with fit summaries for Home
POST	/chat	Send a query; streams the reply (server-sent events)
GET	/recipes/{id}?servings=	Recipe detail with per-member fit and scaled ingredients
GET / PUT	/menus?date=&slot=	Read or update the day menu
POST	/menus/{id}/confirm	Confirm menu and create the ticket
GET	/tickets?date=&status=	Cook's and family's ticket lists
PATCH	/tickets/{id}	Change status, reassign, edit, cancel
POST	/tickets/{id}/stock-issue	Report an out-of-stock item; returns AI suggestions
POST	/tickets/{id}/resolve	Accept a suggested recipe, swap or new dish
POST	/tickets/{id}/messages	Cook ↔ family messages on a ticket
GET / PATCH	/pantry, /pantry/{staple_id}	Read and update stock levels
GET / POST / PATCH	/ration-requests, /ration-requests/{id}	Raise, edit, approve, mark bought
GET	/ration-requests/{id}/bill, /menus/{id}/bill	Bill-style list for manual ordering (JSON + shareable image)
POST	/tickets/{id}/ratings	Submit ratings for a completed ticket
GET	/cooks/{id}/ratings	Ratings feed and weekly summary for the cook

9.2 External services

Service	Used for	Notes
LLM API	Query parsing, reply writing, out-of-stock alternates	JSON-schema output for parsing; temperature 0.3 for replies
Embedding + vector search	Keyword and semantic recipe search	Index recipe name, tags, ingredients
SMS OTP provider	Login	India DLT-compliant sender
Push notifications (FCM / APNs)	New tickets, stock issues, rating prompts, off-duty alerts	Per-role notification settings
Google Play Billing, App Store subscriptions	7-day trial, then ₹50/week	Server-side receipt validation
WhatsApp share intent	Sharing bill-style lists and menus; fallback for a cook without the app	Device share sheet; no WhatsApp Business API in v1
Product analytics	Events in section 12	No personal food data in event names

10. Non-functional requirements

Area	Requirement
Performance	Home ready in under 1.5 s on a mid-range Android phone over 4G; animations hold 60 fps
Motion budget	At most one continuously animated 3D element on screen at a time; all animations use transform/opacity only; honour the OS reduce-motion setting
Images	Serve WebP, 440 px for tiles and 1080 px for detail; lazy-load below the fold; show a cream placeholder while loading
Offline	Last Home list, open menu and grocery list readable offline; chat shows "net nahi hai" and queues nothing

Area	Requirement
Language	UI copy in Hinglish (default), English and Hindi; all strings externalised
Accessibility	Touch targets at least 44 px; text contrast at least 4.5:1; every icon button has a label; screen-reader order matches visual order
Privacy	Diet, allergy and health-like tags are personal data: encrypted at rest, never sold, deletable with the household
Reliability	Chat falls back to the rule-based parser if the LLM fails or times out after 8 s
Platforms	iOS 16+ and Android 10+; phone portrait only in v1

11. Design system and motion

Build the component library in three tiers, as on the "Design library" artboard: Constants (tokens) → Elements (atoms) → Components. Screens use components only, never raw values.

11.1 Colour tokens

Token	Hex	Use
haldi	#FFCC33	Headers, brand surfaces
dhaniya	#0E7C3A	Primary buttons, ADD, selected states
dhaniya-deep	#0E5F31	Hero banner, pressed ledge (#095A29)
tawa	#1B1B18	Body text, user chat bubble
mirchi	#C4331F	Spice indicators, errors
chaawal	#F4F5EF	App background
malai	#F7F3E6	Photo placeholders
fit-soft / tweak-soft / no-soft	#E7F4EA / #FFF1C7 / #FCE6E2	Fit badges

11.2 Type, spacing, radius

- Display and titles: Bricolage Grotesque 800 (40 / 26 px). UI and body: Figtree 500–800 (16 / 14 / 12 px).
- Spacing on a 4-pt scale: 4, 8, 12, 16, 24, 32, 48.
- Radius: 6 badge, 10 segment, 14 button, 16 card, 20 banner, full for chips.

11.3 Key components

Recipe tile, member card, chip, segmented control, ADD button (outline → filled when added), veg/egg/non-veg mark, fit badge, chat bubbles (user / Maharaj), recipe carousel card, sticky menu bar, bottom nav with raised Maharaj button, toast.

11.4 Motion

- 3D thali on Welcome, Loading and the Home banner: the real thali photo on a plane tilted 62°, spinning once per 18 s, floating 10 px on a 4.5 s loop, with rising steam.
- Screen entry: 18 px rise + fade, 450 ms. Recipe tiles pop in with a 38° tilt, staggered 70 ms. Chat cards flip in from 65° on the Y axis.
- Primary buttons sit on a 4 px ledge that collapses on press. Typing indicator = three flipping rotis.
- Recipe detail photo slowly zooms 2% → 14% over 14 s.

12. Analytics and success metrics

Track these events from day one; targets are set after the first 4 weeks of beta data.

Event	Fired when	Properties
onboarding_completed	Household reaches Home the first time	member_count, seconds_taken, used_demo, cook_setup
trial_started / subscribed / trial_expired	Subscription state changes	day of trial, store
query_sent	Chat message or quick ask sent	source (chip / typed / search), slot, language
recommendation_shown	Home grid or chat cards rendered	slot, recipe_ids, fit_badges
recipe_opened	Recipe detail opened	origin screen, rank position, role

Event	Fired when	Properties
dish_added	ADD tapped	origin screen, fit_badge
ticket_created	Menu confirmed	dish_count, minutes_before_due
ticket_status_changed	Any ticket state change	from, to, actor role
cook_duty_changed	Cook goes on or off duty, or someone claims Chef for the day	type, scheduled (bool)
stock_issue_raised	Item marked Out on a ticket	staple category, resolution step (alt recipe / swap / new dish)
ration_request_sent	Cook or admin sends a request	item_count, auto_filled_share
bill_generated	Bill-style list created or shared	source (menu / ration), item_count, shared (bool)
rating_submitted	A member rates a dish	stars, tag_count, on_behalf

Metrics to watch: onboarding completion rate, share of days with at least one menu shared, queries per menu shared, share of shared menus where every member is covered, tickets completed on time, stock issues resolved without a dish change, and the cook's average rating.

13. Decisions, assumptions and open questions

Assumptions made in this document

- Native mobile app first; no web app in v1.
- Cooking is a role: most often mom or another family member, sometimes a hired cook, often both. WhatsApp is only a fallback for a hired cook without a smartphone.
- Recipes come from a curated, licensed database, not live web scraping.
- Meal-slot cut-off times (11:00 and 16:00) are defaults and may need tuning.

Decisions taken

Question	Decision
Recipe source	At least 1,000 recipes curated from publicly available sources, covering every Indian regional cuisine, seen from a home-kitchen angle (section 7.4). No paid content partner in v1
Grocery ordering	No grocery partner. "Order missing items" and approved ration requests produce a bill-style list; the household admin orders manually
Pricing	7-day free trial, then ₹50 per week for all v1 features. Pricing will change with later upgrades
Learning from meals	Yes. Maharaj uses ratings and what was actually cooked to re-rank future suggestions (FR-F5)
Cooks working in several homes	Phase 2. v1 onboards families; the cook is only the executor inside one household
Rewards for the cook	Dropped from v1. Cooks see ratings only
Voice-only cook mode	Not needed in v1

Still open

- [ ] Do we need Hindi-script recipe steps for cooks who don't read English?